

ARA-290

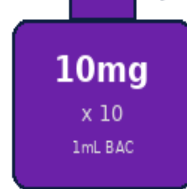
Erythropoietin-Derived Peptide | 10mg x 10 Vials

Neuroprotection | Anti-Inflammatory | Neuropathy Research | SubQ

Purity: 99.8% | Endotoxin-Free | Research Use Only

ARA-290 is an 11-amino acid cyclic peptide derived from the helical structure of erythropoietin (EPO), specifically designed to activate the innate repair receptor (IRR / beta-common receptor) without erythropoietic effects. It provides the tissue-protective and anti-inflammatory benefits of EPO without stimulating red blood cell production. Key applications: diabetic neuropathy, sarcoidosis-related neuropathy, corneal nerve regeneration, and inflammatory disease.

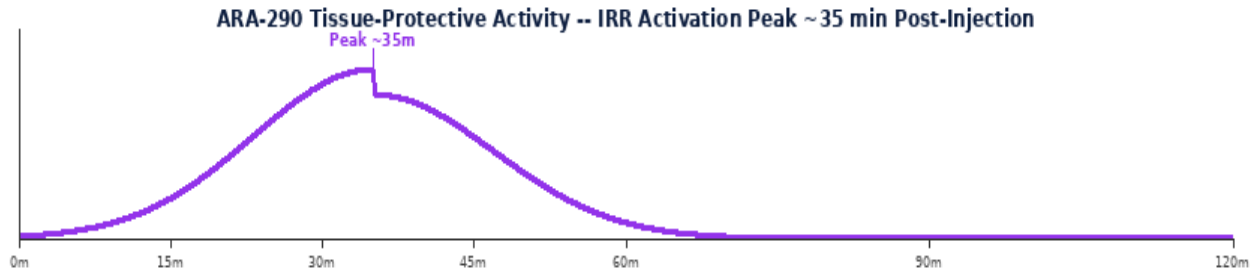
ARA-290 Vial Comparison



10mg
10.0mg/mL

Parameter	Specification
Class	EPO-derived cyclic peptide -- innate repair receptor (IRR) agonist
Vial Size	10mg x 10 vials Purity 99.8%
Reconstitution	1 mL BAC Water -> 10mg/mL (10,000 mcg/mL)
Standard Dose	1,000-8,000 mcg (1-8mg) per injection SubQ
Frequency	1x daily SubQ no fasting required
Key Advantage	All tissue-protective EPO benefits WITHOUT erythropoiesis or thrombosis risk
Cycle & Storage	4-12 weeks on, 4 weeks off 2-8 deg C use within 28 days

MECHANISM OF ACTION & RESEARCH BENEFITS



Pathway	Mechanism	Benefit
IRR Activation (non-erythropoietic)	Selectively binds innate repair receptor (beta-common receptor / IRR); activates tissue-protective signalling pathways (PI3K/Akt, JAK2/STAT5); does NOT bind classical EPO receptor (EpoR homodimer); zero erythropoietic activity at any dose	Tissue protection without haematological risk; safe for non-anaemic subjects
Neuroprotection & Nerve Repair	Promotes peripheral nerve regeneration; reduces axonal degeneration; anti-apoptotic in dorsal root ganglion neurons; clinical evidence in diabetic and sarcoidosis-related small-fibre neuropathy; corneal nerve regeneration demonstrated	Diabetic neuropathy; small-fibre neuropathy; nerve regeneration research
Anti-Inflammatory & Repair	Reduces NF-kB and pro-inflammatory cytokines (TNF-alpha, IL-6); activates anti-apoptotic BCL-2 pathway; promotes tissue repair in heart, lung, kidney, and neural tissue; reduces ischaemia-reperfusion injury in multiple organ systems	Multi-organ protection; ischaemia; inflammatory disease research

Benefit	Context
No Erythropoietic Risk	ARA-290 activates only the innate repair receptor, not the classical EPO receptor -- zero red blood cell stimulation, zero thrombosis risk.
Clinical Evidence	Phase II trials demonstrated significant improvements in diabetic small-fibre neuropathy and sarcoidosis neuropathy -- corneal nerve density improvement confirmed by confocal microscopy.
Broad Organ Protection	Tissue-protective effects demonstrated in heart, kidney, lung, retina, and peripheral nerve -- a uniquely versatile repair peptide.

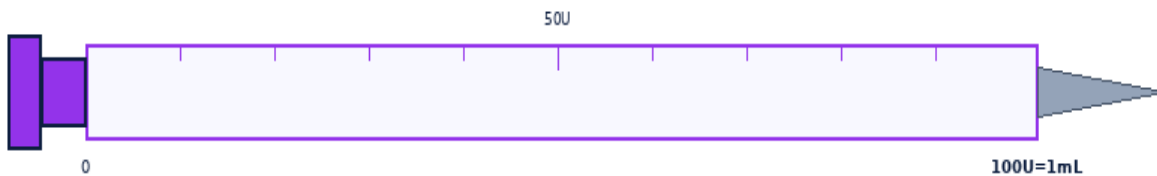
RECONSTITUTION & DOSE CALCULATION

All vials use 1 mL BAC Water. Room temperature. Swirl gently -- never shake.



1 mL BAC Water per vial. Roll gently -- NEVER shake.

Step	Action	Detail
1	Gather	Vial, BAC Water, insulin syringe (29-31G), swabs, sharps.
2	Clean stoppers	Wipe both stoppers. Air-dry 10-15 sec.
3	Draw 1mL BAC	Exactly 1 mL. Expel air bubbles.
4	Inject into vial	Aim at GLASS WALL. Slowly. Never onto powder.
5	Swirl & label	Roll 20-30 sec. Clear within 60 sec. Never shake. Label date. 2-8 deg C. Use within 28 days.



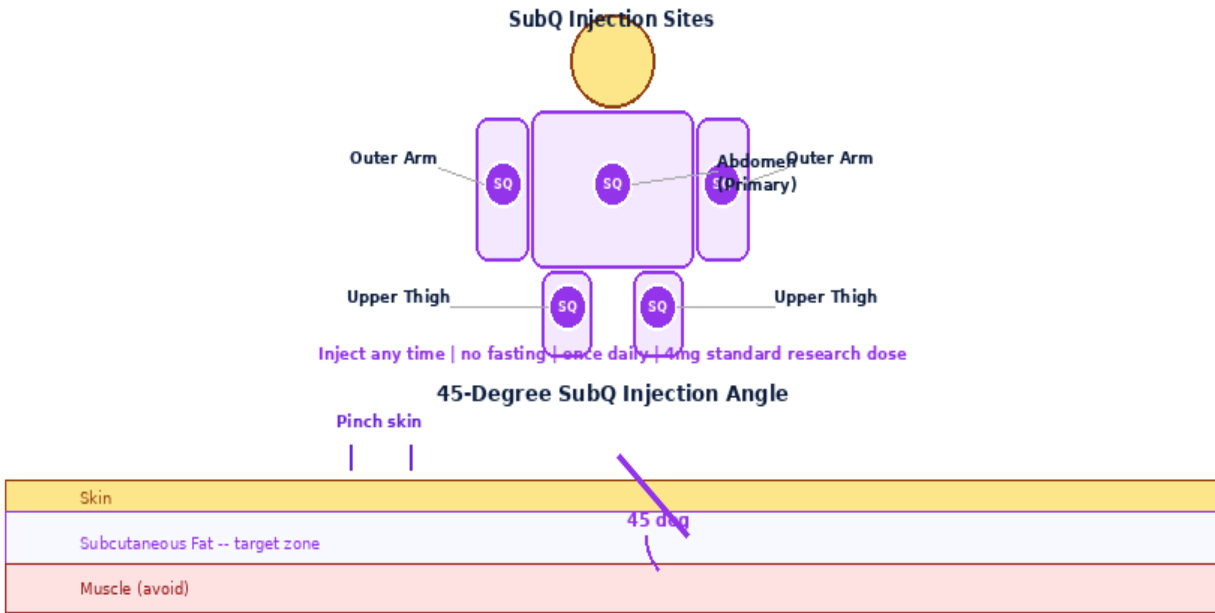
1mL Insulin Syringe | 29-31G | SubQ

Formula: $\text{Vol(mL)} = \frac{\text{Dose(mcg)}}{\text{Conc(mcg/mL)}}$ | **Units:** $\text{Vol} \times 100$

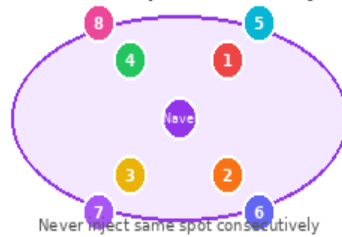
10mg x 10 -- 1mL BAC -> 10.0mg/mL (10,000mcg/mL)

Dose(mcg)	1000mcg	2000mcg	4000mcg	8000mcg
Vol(mL)	0.1000	0.2000	0.4000	0.8000
Units	10.0U	20.0U	40.0U	80.0U
Doses/Vial	10	5	2	1

INJECTION TECHNIQUE, PROTOCOLS & GOLDEN RULES



8-Point Rotation Map -- Rotate Every Injection



Step	Action	Detail
1	Wash & prepare	Wash hands. 29-31G insulin syringe. No fasting required.
2	Draw dose	Standard: 4,000 mcg (4mg). Draw exact volume per dose table.
3	Inject SubQ	Abdomen preferred. Pinch 2-3cm. 45 deg. Inject slowly. Rotate sites.
4	Rotate & dispose	8-point rotation. Press site 5 sec. Sharps container.

Protocol	Dose	Frequency	Duration	Notes
Neuropathy Research	4,000 mcg (4mg)	1x daily	4-8 wk 4 off	Clinical trial dose range. Main research protocol.
Starting Dose	1,000-2,000 mcg	1x daily	4 wk then escalate	Start low to assess tolerance.
High-Dose Research	6,000-8,000 mcg	1x daily	4-8 wk	Higher dose for severe neuropathy models.

Side Effect	Likelihood	Management
Injection site reaction	Common	Mild redness. Rotate sites. Fresh needle.
Mild headache	Occasional	Dose-related. Reduce dose. Ensure hydration.
Fatigue	Occasional	Initial adaptation. Resolves within 1-2 weeks.

No erythropoiesis	Confirmed absent	ARA-290 does NOT stimulate red blood cell production.
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Contraindications: Cancer history, pregnancy, under 18.

1. 1 mL BAC Water -> 10mg/mL (10,000 mcg/mL).
2. No fasting required. Inject any time -- once daily.
3. ARA-290 activates only the innate repair receptor -- ZERO erythropoietic activity.
4. Standard research dose: 4mg (4,000 mcg) once daily SubQ.
5. Rotate injection sites through the 8-point abdomen map.
6. Clinical evidence: significant improvement in diabetic small-fibre neuropathy.
7. Corneal nerve regeneration confirmed by confocal microscopy in Phase II trials.
8. Cycle 4-12 weeks on, 4 weeks off. Store 2-8 deg C, use within 28 days.
9. Contraindicated in known EPO-derived peptide hypersensitivity. Not for pregnancy or under 18.

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